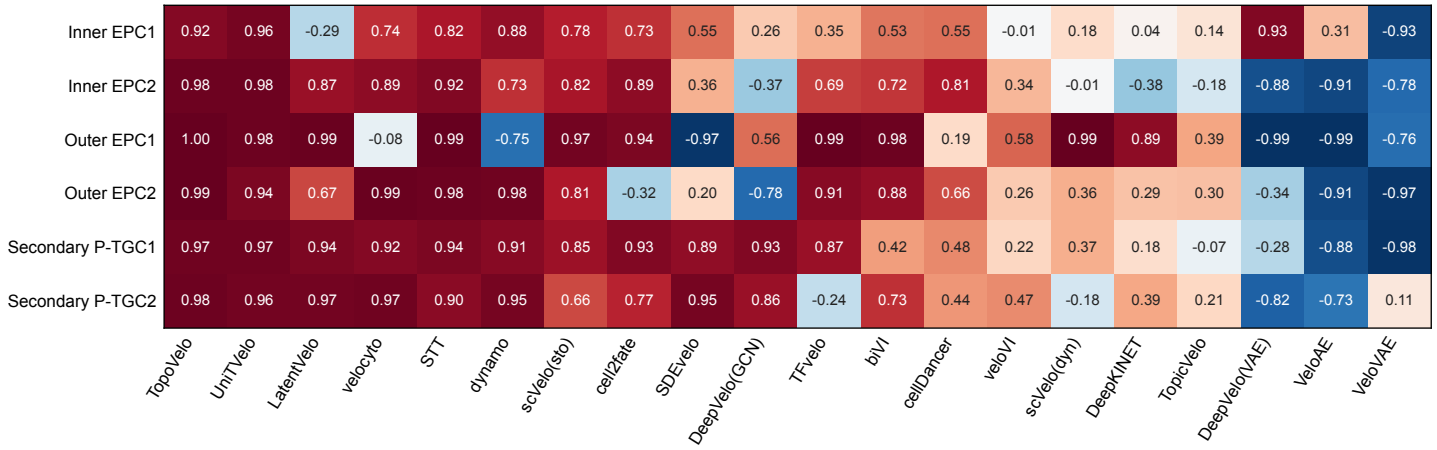
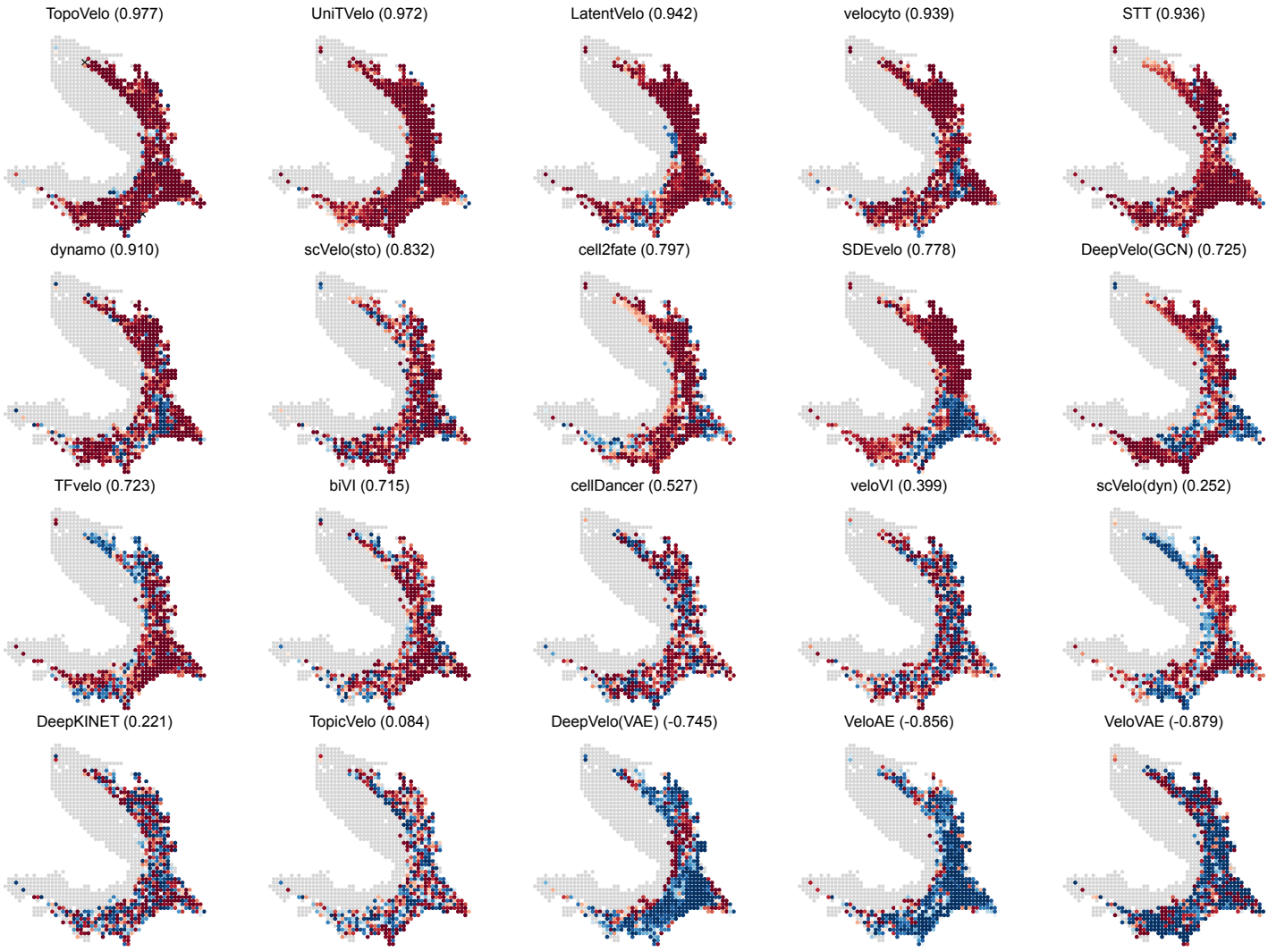
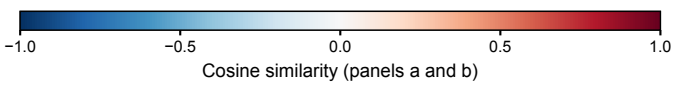
a Spatial direction consistency by cell type



b Spatial distribution of paired direction consistency



Grey: non-target spots; black crosses: undefined pairs. Medians in parentheses.



Extended Data Fig. 10 | Spatial direction consistency with and without non-target spots. Paired spatial-direction consistency for the same 608 target spots in the 1,292-spot ST_raw dataset and the 608-spot ST_small subset. **a,** Median spatial cosine by cell type, with six rows and twenty method columns: Inner EPC1 (47 spots), Inner EPC2 (87), Outer EPC1 (64), Outer EPC2 (61), Secondary P-TGC1 (153), and Secondary P-TGC2 (196). **b,** Per-spot spatial cosines without spatial smoothing; the other 684 spots are grey, and medians appear in parentheses. Red denotes positive cosine (same direction at +1), white denotes zero (perpendicular), and blue denotes negative cosine (opposite at -1) in both panels. Methods retain the descending spatial-median order. TopoVelo has 606 valid spatial pairs and the other 19 methods each have 608; the two TopoVelo pairs with zero velocities are undefined and shown as black crosses in b. TopicVelo spatial directions use native transition matrices, integrating eleven and five usable topics for ST_raw and ST_small, respectively. Spot identifiers and annotations were matched. The across-method boxplot summary is shown in Fig. 5c; Fig. 5b shows selected maps. Preprocessing, retained genes, neighborhoods and fitted models differ between the two saved runs. This is a descriptive comparison of saved-output sensitivity, not a controlled estimate of the isolated effect of non-target spots. Spots are observations from one supplied dataset, not independent biological replicates. Consistency does not establish directional accuracy or velocity magnitude.